SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING - CSA17 ARTIFICIAL INTELLIGENCE LAB

Day 1 - Program 4	Course Outcome: CO2	Language: Python 3

Question

Write the python program for Crypt-Arithmetic problem

Aim

To solve the cryptarithmetic puzzle SEND + MORE = MONEY by assigning a distinct digit to every letter.

Algorithm / Explanation

Each letter must stand for a different digit 0-9 and the first letter of a word cannot be 0.

All permutations of digits for the 8 distinct letters are tried (generate and test) until SEND + MORE = MONEY holds.

Program

```
# Cryptarithmetic Problem:  SEND + MORE = MONEY
from itertools import permutations

def solve(word1, word2, result):
    letters = sorted(set(word1 + word2 + result))
    if len(letters) > 10:
        return None
    first = {word1[0], word2[0], result[0]}          # leading letters cannot be 0

    def value(word, m):
        n = 0
        for ch in word:
            n = n * 10 + m[ch]
        return n

    for perm in permutations(range(10), len(letters)):
        m = dict(zip(letters, perm))
        if any(m[ch] == 0 for ch in first):
            continue
        if value(word1, m) + value(word2, m) == value(result, m):
            return m
    return None

sol = solve("SEND", "MORE", "MONEY")
if sol:
    print("Letter assignment:", sol)
    s = lambda w: "".join(str(sol[c]) for c in w)
    print(f"  {s('SEND')}\n+ {s('MORE')}\n------\n {s('MONEY')}")
else:
    print("No solution found")
```

Output

```
Letter assignment: {'D': 7, 'E': 5, 'M': 1, 'N': 6, 'O': 0, 'R': 8, 'S': 9, 'Y': 2}
  9567
+ 1085
------
 10652
```

Result

Thus the Python program was executed successfully and the output was verified.